

The Snow Energy Balance During Extreme Ablation: A Case Study of the March 2026 Heatwave in the Sierra Nevada

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Key Points:

- Record heat coincided with record rates of snow ablation during March of 2026, leading to record low-to-no-snowpack conditions.
- Net-shortwave was the dominant ablation driver despite the anomalously warm air temperatures.
- Low snow-albedo and higher than average downwelling shortwave due to a lack of mid level clouds contributed to record rates of shortwave heating of the snowpack.
- Snow ablated more slowly under vegetation canopy due to enhanced solar shading.
- Less than **TODO: x** of total snow ablation was due to sublimation.
- Boundary layer decoupling and vapor pressure arguments explain the small contribution of sensible heating and sublimation to the snow energy and mass balance during the heatwaves.

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Abstract

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1 Introduction

While meltwater from seasonal snowpacks has historically provided $\sim 70\%$ of runoff in the western USA (Li et al., 2017), these natural reservoirs and our ability to measure, monitor, and predict them are in decline (Hale et al., 2023; Cowherd et al., 2024). From a practical perspective, many operational snowmelt runoff models use a temperature index approach, whereby air temperature (T_{air}) is the primary predictor of melt (Anderson, 1973). While these models have proven effective, the physical justification for such models is not clear. Since at least the 1980s the field of snow hydrology appreciates that understanding the climatic behavior of snow necessitates considering the entire energy budget of the snowpack. In particular, net shortwave radiation incident on the snowpack (SW_{net}) is a first-order predictor of rates of melt, and it is well accepted that both small impurities (Deems et al., 2013) and natural grain-growth mechanisms that decrease the amount of reflected sunlight (the snow albedo) are responsible, in conjunction with rising sun-angles in the spring, for the majority of energy available for snowmelt. Unlike downwelling longwave ($LW \downarrow$; ~~4-1004-100~~ μm), shortwave ($SW \downarrow$; ~~0.3-3-3~~ μm) is mainly insensitive to variations in T_{air} itself. Variations in $SW \downarrow$ in time (and across elevation) are driven primarily by Rayleigh scattering by O_2 and N_2 , Mie scattering by natural occurring and human-made aerosols, dust, and cloud droplets, as well as absorption primarily by water vapor, CO_2 , and O_3 .

At the same time, recent work has highlighted the strong correlation between extreme snow ablation (the loss of snow to either sublimation or melt) events and so-called “snow-eater heatwaves” — prolonged periods of sustained (day and night), anomalous, and above freezing T_{air} during the snow season (Rhoades et al., 2026). The characteristics of snow-eater heatwaves have already appreciably changed since the 1850s. These events are occurring more frequently, at a higher intensity, over a larger portion of the western US (WUS), and earlier in the water year.

An example of such an early-season, high-intensity, snow-eater heatwave (SEHW) occurred in March of 2026, when the monthly March T_{air} across the WUS were 3 standard deviations above long-term averages observed at many stations. This event (which started earlier than the window of occurrence studied in Rhoades et al. (2026)) resulted in widespread and rapid ablation at many snow-observing sites and one-to-two month earlier than average streamflow in many watersheds. This raises an apparent paradox: if $SW \downarrow$ provides most of the energy for snowmelt, and yet it is largely insensitive to T_{air} , then why are heatwaves so efficient at ablating snow?

Using the March 2026 event as a case study, we ask the following questions:

- What are the dominant snowmelt energy forcings during early-season heatwaves, and how do they scale with increasing T_{air} ?
- How do vegetation canopies and atmospheric conditions — including clouds or lack thereof — shape such energetic forcings during heatwave events?
- What are the relative contributions of sublimation and melt to extreme snow ablation during such events?

For some perspective, approximately half of North America’s snow water volume resides under tree canopy (Kim et al., 2021). Canopies ~~absorbs~~absorb shortwave radiation, ~~increases the~~increase longwave radiation, and ~~reduces~~reduce winds relative to open or non-vegetated snow surfaces, ~~however~~however, the magnitudes of these changes can vary widely across space and time. A number of studies have considered the effects of vegetation and forest treatments on the mean-state of the snow climate of this region and it is well observed that snow tends to accumulate more and ablate ~~layer~~later in the year in open regions relative to the sub-canopy (Harpold et al., 2020; Krogh et al., 2020), and synthesis work has shown that open versus subcanopy snow dynamics are dependent on latitude and mean winter T_{air} (Lundquist et al., 2013). However, the interactions between vegetation and discrete extreme heatwave events has not previously been considered.

While low clouds have the most substantive effect on the snow energy balance (Rudisill et al., 2025), despite otherwise drier-than-normal conditions, ~~high-altitude~~high-altitude cirrus clouds can occur during high-pressure ridging events commonly associated with cool-season heatwaves (Rhoades et al. (2026) Figure 1 and Supplemental Figure S5). Cirrus increases $LW \downarrow$ and decrease $SW \downarrow$ relative to clear skies (L’Ecuyer et al., 2019), though the radiative impacts to snowpacks are underexplored in the literature, let alone during heatwave events.

To answer these questions, we use the March 2026 event as ~~an case study~~a case study to explore heatwave-snow interactions and place the record heat and associated snow ablation in a climatic and synoptic context. We focus our analysis on the greater Lake Tahoe region of the Sierra Nevada region and use multiple lines of evidence (reanalyses, long-term station records, remote sensing observations, and models) to understand the energetic components of the anomalous observed snow ablation.

The outcomes from this study answer basic science questions about energy available for snowmelt or sublimation in water-resource critical basins in the Sierra Nevada, with implications for other regions. The findings can directly inform efforts towards management and resilience against extreme events by providing actionable steps towards improving the monitoring and prediction of snowmelt. Last, improved understanding of vegetation and snow interactions during heatwaves can inform headwater management practices through improved decision support tools (e.g., Heggli et al. (2026)).

2 Background and Methods

2.1 Study area

We focus our analysis on three basins in the immediate vicinity of the Central Sierra Snow Laboratory (CSSL; 39.32 °N, 120.37 °W, 2100 m.a.s.l) located in Soda Springs, CA (Schwartz et al., 2026). These include the Yuba, North Fork American, and Truckee River Watersheds (the greater Truckee River Watershed is composed of the Lake Tahoe basin and Truckee River basin at the eight-digit hydrologic unit code (HUC8) scale) (**Fig. 1a–d**). The Yuba and North Fork American rivers support the New Bullards Bar and Folsom reservoirs, respectively. These reservoirs have combined storage capacities approaching 2 million acre feet and hydropower generation potential of 559 megawatts **TODO: citation**. All three rivers support varied downstream consumptive uses (i.e., urban, industrial and agricultural) and provide critical aquatic and riparian habitats.

2.2 Observational data

We use data from twelve Natural Resources Conservation Service (NRCS) SNOW-pack TELemetry (SNOTEL) Network stations located throughout the study region spanning **TODO: X** to **TODO: Y** m.a.s.l. SNOTEL is the primary snow observing program

in the ~~western USA~~ WUS (Serreze et al., 1999) and routinely measures snow water equivalent (SWE) and T_{air} , though the homogeneity of T_{air} measurements ~~remain~~ remains a challenge for the SNOTEL network (Oyler et al., 2015; McAfee et al., 2019). To supplement SNOTEL temperature records, we use data from four US Historical Climate Reference Network (USHCRN) stations and the gridded nClimGrid product throughout the study region. USHCRN stations are high-quality stations subset from the larger GHCN-D dataset Menne et al. (2012). The HCNv2 dataset provides homogenized and quality-assured monthly T_{air} based on USHCRN data that dates as far back as the late 1800s (Menne et al., 2009). While this data does not have the same spatial density and elevation range as SNOTEL, the longer temporal record and high quality control mean that these observations can more confidently be used for evaluation of climate extremes, such as heatwave events (Bercos-Hickey et al., 2022).

In addition to hosting an NRCS SNOTEL site, CSSL provides long-term meteorological and snowpack records in addition to surface energy balance measurements. Campbell Scientific CNR4 four-component radiometers measure incoming and outgoing SW and LW radiation from the atmosphere and snowpack. A Campbell Scientific IRGASON eddy covariance system measures sensible (H_s) and latent (H_l) heat fluxes above the snowpack. Both the IRGASON and CNR4 are hosted on a movable arm (Fig. 1c) that automatically adjusts to maintain a constant distance of approximately one meter above the snow surface as snow accumulates or ablates throughout the season. Eddy covariance fluxes were processed using the Licor EddyPro v7.0.9 software with despiking, double coordinate rotation, and flux quality assignment. ~~TODO: X%~~ of the H_s and H_l fluxes are retained for analysis during the month of March.

The CSSL forest study site hosts an additional meteorological station approximately 150 m ~~North~~ north of the main CSSL site. This tower hosts an additional Kipp and Zonen SP Lite2 radiometer that measures below-canopy up and downwelling shortwave radiation. SWE measurements were obtained using a federal sampler (Osterhuber, 2014) at nine discrete sampling intervals during the March heatwave in the forest location.

2.3 Atmospheric structure and radiation

To investigate the synoptic patterns associated with the March heatwave, we use 500 hPa geopotential height and atmospheric column information from the fifth-generation ECMWF Reanalysis (ERA5~~reanalysis~~) (Hersbach et al., 2020). Daily geopotential height anomalies during the event are computed using data provided by Lee et al. (2023), which has been de-trended and ~~covers~~ covers the period from 1979 to 2022.

We use the Clouds and the Earth's Radiant Energy System (CERES) EBAF and SYN1Deg products, hereafter CERES-EBAF (Kato et al., 2018) and CERES-SYN1Deg (Rutan et al., 2015)~~products~~, to investigate the effects of clouds or lack thereof on the snow energy balance. CERES-EBAF provides ~~the gold-standard of measurement of surface and TOA measurements of surface and top-of-atmosphere (TOA)~~ irradiances at a 1x1 degree resolution at a monthly cadence. This product is suitable for ~~trend analysis~~ trend analysis. CERES-SYN1Deg is a ~~lower-level~~ lower-level product that provides hourly estimates of the same quantities but is not configured to examine long-term temporal trends. Thus, we use CERES-SYN1Deg all-sky and clear-sky $SW \downarrow$ and $LW \downarrow$ (e.g., hypothetical irradiance in the absence of clouds) to compute cloud-radiative effects following Ramanathan et al. (1989) and subsequent studies. Rudisill et al. (2025) compared cloud radiative effects from CERES-SYN1Deg against in-situ radiometers in the Colorado Rockies and found good agreement, despite grid-resolution mismatches and confounding effects of terrain. Hinkelman et al. (2015) likewise found CERES-SYN1Deg proved efficacious when used as forcings for snow simulations. The ~~2000-present~~ 2000-present period-of-record of CERES-EBAF ~~are~~ is used to examine the March anomaly in $SW \downarrow$ and $LW \downarrow$ and the contribution of clouds to the anomalies therein.

2.4 SUMMA model configuration

To investigate the snow energy balance, we use the [SUMMA Structure for Unifying Multiple Modeling Alternatives \(SUMMA\)](#) model (Clark et al., 2015). SUMMA is well-vetted for a range of snow climate applications and provides several options for flux stability corrections, canopy radiation interactions, snow-layering schemes, and other model structural options (Cristea et al., 2022). We use the Jordan snow layering option, which allows for up to 100 snow layers, the UEB-2 stream approximation for canopy radiation interactions, the “Louisinv²” option for stability correction of turbulent fluxes, and the “stickySnow²” canopy snow interception scheme. SUMMA solves for a vegetation energy balance and canopy air-space temperature which is used to produce $LW \downarrow$ incident onto the sub-canopy snowpack. Additional specific modeling parameterizations with relevance to this study are described in Sec. 2.5 and the Supplemental Material. We assign a measurement height of 12m for the model forcings into ~~the~~ SUMMA in order to intercompare the effects of 10m tall needleleaf vegetation characteristic of the forest site at CSSL using the same forcing variables.

We use hourly ~~data from the fifth-generation ECMWF (ERA5)-reanalysis data~~ (Hersbach et al., 2020) as SUMMA model forcings after adjusting to the CSSL site. ~~*~~ Daily precipitation from the NRCS SNOTEL site is used directly as forcing when available. Hourly ERA5 two-meter T_{air} are bias corrected to match the monthly average T_{air} observed at the CSSL from the NRCS records by removing a constant linear bias term. Comparing ERA5 $SW \downarrow$ data to observations from the CNR4 shows good agreement except for morning and evening when the surrounding vegetation blocks direct beam solar radiation. We compute and apply a $SW \downarrow$ correction coefficient as a function of the solar zenith and azimuth angles to adjust the ERA5 data to account for shadows ([See see](#) Figure S3). To ensure that the ERA5 data matches the best available estimates of variability and trends in radiation at this site, we use a multiplicative scale that forces the ERA5 $SW \downarrow$ monthly anomalies to match monthly anomalies in the CERES-EBAF data. Humidity, wind, and $LW \downarrow$ variables validated well against the available observations and are not adjusted. Additional information about SUMMA validation is provided in the ~~supplemental material (Supplemental Material , Section~~ [Supplemental Material \(Sec. 3\)](#).

2.5 The snow energy balance

Using meteorological sign conventions for turbulent fluxes, the energy budget of a snowpack is given by:

$$\text{Melt/Refreeze} + \Delta CC = \underbrace{H_s + H_l}_{\text{turbulent fluxes}} + \underbrace{LW \downarrow - \sigma \epsilon T_{snow}^4}_{\text{net longwave}} + \underbrace{SW \downarrow (1 - \alpha)}_{\text{net shortwave}} + G \quad (1)$$

where Melt/Refreeze is the rate of melt or liquid water refreezing in mass per second multiplied by the latent heat of fusion, ΔCC is the time rate-of-change in the snowpack cold-content (Jennings et al., 2018), H_s and H_l are the turbulent fluxes of sensible and latent heat respectively at the snow/atmosphere interface, T_{air} is the temperature of the air, T_{snow} is the temperature of the snow surface, ϵ is the emissivity of the snow (0.98), $SW \downarrow$ is the downwelling shortwave radiation, and α is the albedo of the snow surface defined as the ratio of upwelling shortwave radiation ($SW \uparrow$) to $SW \downarrow$.

The mass-balance of the snowpack is related to the energy balance (Eq. 1) in the following ways. The daily or monthly change in SWE is related to both Melt and to a lesser extent H_l , though the latter may remove 10% to upwards of 50% of the seasonal snow mass depending on a variety of factors (Sextstone et al., 2018; Schwat et al., 2025). Melt may be either positive (melt) or negative (refreezing) using the sign convention in Eq. 1. Melt integrated throughout an entire day is not synonymous with a loss in SWE,

since meltwater may be retained in the snowpack and refreeze. For flow through the bottom boundary of the snowpack, the dry pore space of the snow must be filled with liquid water to residual saturation (Colbeck, 1976; Seligman et al., 2014). Snowpacks may retain between ~~2-82~~2-8% liquid water by volume. In the SUMMA model, meltwater flow throughout the snowpack layers is modeled as a gravity-driven power law; preferential flow is not considered.

2.5.1 The influence of canopy

Vegetation canopy influences the snow energy balance through several mechanisms. First, falling snow is intercepted by the canopy and may melt or sublimate from the canopy before reaching the snowpack, so total SWE accumulation under canopy is commonly less than adjacent open areas (Musselman et al., 2008). Second, canopy layers may inhibit turbulent exchange of mass and energy by suppressing wind. Third, vegetation impacts the radiative balance of the snowpack by altering $SW \downarrow$ and $LW \downarrow$ impacting the snow surface. In SUMMA, the ~~the~~ incident $LW \downarrow$ to the sub-canopy snowpack is modelled by:

$$LW \downarrow_{\text{under canopy}} = \underbrace{(1 - \epsilon_c) LW \downarrow_{\text{atm}}}_{\text{sky LW through canopy gaps}} + \underbrace{\epsilon_c \sigma T_c^4}_{\text{canopy emission}} \quad (2)$$

, where ϵ_c is the emissivity of the canopy layer. In SUMMA, the ϵ_c is purely a function of the ~~leaf~~leaf-area index (LAI) and stem-area ~~indices~~index. We turn ~~of-off~~ vegetation phenology in the model such that the ϵ_c is constant for these experiments. For $SW \downarrow$ below the canopy, we use the “UEB-2stream” SUMMA parameterization which is based on Mahat and Tarboton (2012). An LAI value of 1.5 was assigned based on data from ~~naerby~~nearby representative regions (Krogh et al., 2020). These choices are further validated by comparing modelled sub-canopy $SW \downarrow$ for a ~~rainge~~range of LAI values against the CSSL radiometers in the forest site, and finding good agreement, supporting these parameterization choices (Fig. S4).

2.5.2 Defining radiative forcing and effect

We apply the radiative forcing concept in several ways in this study (Ramanathan et al., 1989; Deems et al., 2013; Rudisill et al., 2025). In this context, radiative forcing quantifies the instantaneous difference between the net or a single stream of radiation between observed conditions and a counterfactual reference state.

We define RF_{canopy} as the difference between R_{net} under the canopy relative to R_{net} outside of the canopy layer. Lacking observations sufficient to constrain RF_{canopy} , we use the SUMMA model to evaluate this quantity and express this value for both LW and SW components. To investigate the relative contribution of α decay on the surface radiation balance, we quantify the α radiative forcing (~~RF $_{\alpha}$~~ RF_{albedo}) by differencing SW_{net} computed using observed α versus SW_{net} computed using observed $SW \downarrow$ and a hypothetical fresh-snow broadband α ; for this work we use 0.85.

We define a cloud radiative “~~effect~~effect”, rather than “~~forcing~~forcing” to express the influence of clouds on downwelling $LW \downarrow$ and $SW \downarrow$ terms only (RE_{cloud}), since a full accounting of cloud radiative effects on net-energy terms also depends on the amount reflected and absorbed by the surface, and therefore the albedo (Rudisill et al., 2025) and assumptions about varying upwelling emissions. All radiative effects are signed such that positive indicates energy is added to the snowpack via that process (clouds, canopy, or α decay), and a negative sign indicates the opposite.

2.6 Ranking SWE and heatwave anomalies

To rank or contextualize SWE anomalies, we use the framework from Hatchett et al. (2022), which uses a percentile-based metric to assign snow drought categories. Drought

Table 1. Definitions of radiative effects used in this study. All effects are signed positive when energy is added to the snowpack. x refers to the radiative component, which is either SW, LW, or net (the sum of SW and LW).

Name	Abbreviation	Equation
Cloud Radiative Forcing <u>Effect</u>	RE_{cloud}	$F_{x,cloudy} \downarrow - F_{x,clear} \downarrow$
Canopy Radiative Forcing	RF_{canopy}	$F_{x, under-canopy} - F_{x, open}$
Albedo Radiative Forcing	$RF_{\bar{\alpha}_{albedo}}$	$SW \downarrow [(1-\alpha_{obs}) - (1-\alpha_{clean})]$

categories are ranked as abnormally dry (D0), moderate drought (D1), severe drought (D2), extreme drought (D3), and exceptional drought (D4) for values between the ~~30th–20th~~, ~~20th–10th~~, ~~10th–5th~~, and ~~5th–2nd~~ 30th–20th, 20th–10th, 10th–5th, and 5th–2nd and below the 2nd percentiles, respectively. SWE percentiles are computed using the NRCS SNOTEL periods of record for each site.

Z-scores are used to rank near-surface T_{air} and snow ablation anomalies, given by

$$z = \frac{X - \mu}{\sigma}$$

. Baseline mean and standard deviations are computed using a common ~~1981–2020 normal periods~~ 1981–2020 normal period unless otherwise stated.

3 Results

3.1 The March 2026 SEHW

A four-station composite of the long-term and homogenized monthly ~~USHCN~~ USHCRN HCNv2-data shows that 2026 had the highest March T_{air} observed since the beginning of the record in 1890, and was +3.6 °C above the 1981–2020 average — a full three sigma above the mean ($z=3$) (Fig. 2a). The nClimGrid likewise shows a +4–6 °C anomaly above average in some locations, compared to the +3.6°C anomaly observed at the four-station ~~USHCN~~ USHCRN composite (Fig. 2b).

This prolonged period of heat resulted from an extreme and persistent ridge of high pressure that gained strength throughout March (Fig. 3a,b), with ~~500mb~~ 500 hPa geopotential height anomalies exceeding two standard deviations and 200m throughout March.

Immediately prior to the March 2026 heatwave, the region experienced three notable periods of weather. While the water-year-to-date precipitation (1 ~~October–28~~ October–28 February) was 100% of ~~1991–2020~~ 1991–2020 normal leading up to March 2026 (not shown), there was a prolonged midwinter dry spell (Hatchett et al., 2023) from 7 January to 9 February during which SWE began ablating (Fig. 3c). Then, between February 14th to 19th, a total of 100 mm of new SWE accumulated at CSSL, the third-highest four-day total since records began in 1971. This snowfall event was followed by anomalously warm T_{air} and a rain-on-snow (ROS) event on the ~~24–25~~ 24–25 of February that brought 91 mm of rainfall, leading to a net decline in SWE (Fig. 3b,c).

The beginning of March started with average to slightly above average T_{air} . As the high pressure ridge grew over the WUS, daily maximum T_{air} peaked at 22.2°C on 19 March and 23.3°C on 20 March at CSSL ~~TODO: report daily avg air temp~~ but remained well-above normal from ~~8–31~~ 8–31 March, with daily T_{air} values at the Tahoe City COOP station yielding anomalies of $z=2$ or greater for four days between the ~~17th–27th~~ 17th–27th. Correspondingly, the 500 ~~mb~~ hPa height anomaly exceeded 200 meters across the WUS on this date, and was greater than two- σ above the detrended mean for this date. The

hemispheric wave pattern on 16 March in particular is characteristic of an amplified North American Dipole pattern (S.-Y. Wang et al., 2014; S.-Y. S. Wang et al., 2015; Singh et al., 2016).

Snow began ablating consistently starting with the February ROS event for both the open and forest sites and continues into the month of March. We find good agreement between both SUMMA and observed SWE (Fig. 3c) and rates of ablation. SWE ablated at an average rate of 7.37 mm/day (7.06 mm/day) between the first of March and the 14th of March, after which it accelerated to a rate of 25.30 mm/day (18.66 mm/day) until the end of the month in the open (forest) until ablating completely in the open by the **TODO: 28th** and the forest site by the **TODO: 24th**.

3.2 SWE anomalies during water year 2026

SWE completely ablated earlier, and at a faster rate than typical for the time period, at northern Sierra SNOTELs (Fig. 4a-c). The beginning of March saw SWE levels at a D1 or D0 (“no drought²”) level across the twelve SNOTEL sites (Fig. 4d). While SWE was nearly 0 mm for ~~the~~ two of the lower-elevation sites (station ID numbers 473 and 809) this was not particularly anomalous for those locations for this date. By the end of March, 9 of the 12 stations reached 0 mm SWE and the remaining sites were in a D3 (“extreme drought²”) drought classification (Fig. 4e). Only 4 of the SNOTEL sites had a precedent for 0 mm of SWE on 31 March. Similarly, the snow covered area reached its lowest ever recorded value on 31 March ~~value during the MODIS during the Moderate Resolution Imaging Spectroradiometer (MODIS)~~ record (see Supplemental Material; Fig. S2). The rate of monthly snow ablation was likewise anomalous at most sites (Fig. 4f). Typically, most SNOTEL sites in this region gain (or remain constant) rather than lose SWE throughout March (Fig. 4b) with the exception of two of the lowest elevation sites. Nine out of 12 sites, including CSSL, measured their highest rate of monthly SWE ablation for the SNOTEL record. The lowest elevation site (473) is the only site without anomalous monthly SWE ablation, in part because the starting value was so close to zero.

3.3 The snow energy balance in the open and forest site

To understand the causes of the anomalous observed ablation described in Sec. 3.2, we now focus on the CSSL site (site ID 428 in Fig. 4 and also depicted in Fig. 3c). These causes, in addition to the different rates of ablation observed at the forest and open sites, can be understood by examining a time series of energy balance components (Eq. 1) and by separating them into nighttime, daytime, and 24-hour averages (Fig. 5a-f, Fig. 6a-f).

H_s and H_l are relatively minor factors compared to radiative fluxes (Fig. 5a,b, Fig. 6a,b). Two periods of significant sublimation occurred (negative H_l) on the 5th to the 7th, which correspond with relatively high wind speed (Fig. 3c) but prior to the onset of the warmest March temperatures. Another significant sublimation event took place on the 24th. The highest rates of H_s occurred between the 3rd through the 10th and again from the 21st to the 26th. A maximum daytime H_s exceeding $80 \text{ W}\cdot\text{m}^{-2}$ took place on the 22nd for both the forest and open sites, just one day after the peak of the March temperatures.

SW_{net} far exceeds the H_s maxima in the open site, however (Fig. 5c, Fig. 6c). Here, SW_{net} increases from daytime values approaching $80 \text{ W}\cdot\text{m}^{-2}$ to values exceeding $240 \text{ W}\cdot\text{m}^{-2}$ by the end of the month (with 24hr averages of approximately $120 \text{ W}\cdot\text{m}^{-2}$). Rates are significantly lower in the forest. SW_{net} increases throughout March, but never exceeds $80 \text{ W}\cdot\text{m}^{-2}$.

In the forest, LW_{net} is positive on average for the entirety of March 2026 except the ~~4th-7th~~ 4th-7th, which also corresponds with the only period of significant overnight cold-content production (Fig. 5d, Fig. 6d). Overnight LW_{net} switches from slight neg-

active ($\approx -8 \text{ W}\cdot\text{m}^{-2}$) to positive or near-zero by the 16th in the forest site. The highest rate of LW_{net} occurs on the 22nd, two days after the peak of the ~~he-mat~~ heatwave event, and exceeds $60 \text{ W}\cdot\text{m}^{-2}$ during the daytime.

LW_{net} in the open site has a completely different sign, meaning that the snow is continuously losing heat via longwave emission, even during the peak of the heatwave in the latter half of March, and on the order of -80 to $-40 \text{ W}\cdot\text{m}^{-2}$.

The previous energy exchanges, in addition to ground heat flux (not shown), cumulatively manifest as energy directed towards breaking/reforming ice crystalline bonds (Melt/Refreeze) and warming ice to the melting point (ΔCC). A significant amount of energy is re-added back to the snowpack during the night by liquid water refreezing (~~6-33~~ 6-33 $\text{W}\cdot\text{m}^{-2}$) in the open site, and to a lesser extent the forest site (Fig. 5e, Fig. 6e). The overnight LW_{net} cooling in the open yields overnight ΔCC that is eroded the following day, prior to the initiation of melt (yielding net-zero averages over a 24 hour period; Fig. 5f). The cumulative overnight production in cold-content becomes small compared to the following day's melt by the 14th of March and represents less than 20% of the total melt energy, indicating that the lack of overnight cold-content production — almost purely driven by the decline in LW_{net} and increase in overnight H_s — yields a snowpack that is primed for ablation the following day in the open site.

The dynamics of ΔCC are different in the ~~forest—forest~~ forest — there is a near-zero amount of overnight cold-content production and very little refreezing. These model results are confirmed by field reports from CSSL, who found ~~melt-refreeze~~ melt-refreeze crusts in the open during March but slush in the forest site even during morning observations, in agreement with these results.

3.4 Radiative forcing by canopy, cloud, and snow ageing

The radiative forcing concept (Sec. 2.5.2) can be used to further interpret daily and monthly variation in the energy fluxes described previously and shed light on the differences between the open and forest site (Fig. 7a-f).

Both $\text{RF}_{\text{albedo}}$ ~~and~~ and $\text{RF}_{\text{canopy}}$ have a demonstrable impact on the energy balance. Snow α started at a relatively low value leading up to the March heatwave (0.6 to 0.7) and declines consistently during the month of March (Fig. 7a). The declines are well-matched by remote sensing and CSSL observations (see Supplemental Material).

$\text{RF}_{\text{albedo}}$ increases at the open site ~~from~~ from as a function of both rising sun-angle and $SW \downarrow$ (not shown) and α decline, reaching a value exceeding $60 \text{ W}\cdot\text{m}^{-2}$ in the open site based on the SUMMA simulation. While there is only a minor difference between the SUMMA model snow α between the forest and open sites, the $\text{RF}_{\text{albedo}}$ in the open exceeds $\text{RF}_{\text{albedo}}$ in the forest site by more than a factor of two by the end of the heatwave due to attenuation of $SW \downarrow$ by the canopy layer (Fig. 7b).

Second in terms of magnitude to $\text{RF}_{\text{albedo}}$, $\text{RF}_{\text{canopy}}$ yields a $+20$ to $-90 \text{ W}\cdot\text{m}^{-2}$ maximum daily rate during the month (Fig. 7c). There are two competing effects. $\text{RF}_{\text{canopy}, LW}$ increases as a function of T_{air} at a rate of $3.3 \text{ W}\cdot\text{m}^{-2}$ per $^{\circ}\text{C}$ of T_{air} , which is visualized by considering the daily relationship for the entire SUMMA output record for the month of March (Fig. 7e). Additional analysis (not shown) indicates that the canopy layer is consistently warmer than the ambient T_{air} during the daytime, which combined with the emissivity relationship (Eq. 2) explains this effect. This effect does not exceed the shading effect of the canopy ($\text{RF}_{\text{canopy}, SW}$) however, even though the canopy attenuates much of the incident sunlight, so $\text{RF}_{\text{canopy}, Net}$ is near-zero and transitions to negative (cooling the snow) during the hottest part of the heatwave when the sun is also the most intense and the α is also lowest.

Clouds — most of which are high-cirrus ice clouds ~~TODO: confirm~~ — yield a modest but significant effect on the energy balance. RE_{cloud} is negative for $SW \downarrow$ (decreasing incident sunlight) but positive for $LW \downarrow$ (increasing energy directed to the snowpack), with significant events happening on the first of the month, the 4th, the 10th, and the ~~21-23rd~~ 21-23rd of the month (Fig. 7d). ~~TODO: show cloud fraction.~~

Ultimately, the cumulative energy transmitted from the atmosphere to the snow attributed to RF_{albedo} is $+101.3$ and $+23.7 \text{ MJ}\cdot\text{m}^{-2}$ for the open and forest site respectively and RF_{canopy} is responsible for a $-95.1 \text{ MJ}\cdot\text{m}^{-2}$ effect (reducing energy incident on the snowpack). RE_{cloud} (SW) has a value of $-49 \text{ MJ}\cdot\text{m}^{-2}$ and RE_{cloud} (LW) has a value of $(+31 \text{ MJ}\cdot\text{m}^{-2})$ cumulatively. A full accounting of the net-effect of RE_{cloud} on the snow energy balance also requires taking into account what is reflected and absorbed by the snow. Assuming a weighted albedo value of 0.6 (and that the ~~CERES-SYN1Deg~~ CERES-SYN1Deg data is applicable at this scale) a coarse estimate of the net-cloud radiative forcing yields $+11 \text{ MJ}\cdot\text{m}^{-2}$ of heating to the snow in the open (Fig. 7f). We do not compute a similar value ~~of~~ for the forest site since the co-variance with the vegetation energy and radiative transfer does not yield a pragmatic solution, but we hypothesize that the magnitude may be similar due to the competing effects from SW and LW mechanisms.

To place values of RE_{cloud} in context, we make comparisons against climatological RE_{cloud} in ~~Section~~ Sec. 3.5.

3.5 Climatological context of the March 2026 snow energy balance

How does the March 2026 energy balance compare to typical Marches at CSSL? We have seen previously that, on average, SWE gains rather than ablates (Fig. 4f) during the month of March.

The daily components of the energy budget can be contextualized by examining cumulative sums, in $\text{MJ}\cdot\text{m}^{-2}$, of the different energy balance terms to quantify their total contributions to ablation (Fig. 8a–g). For some perspective, ~~it takes~~ $1 \text{ MJ}\cdot\text{m}^{-2}$ of energy is equivalent to approximately 3.0 mm of equivalent snowmelt or 0.35 mm of snow sublimation (found by dividing by the latent heat of fusion and sublimation, respectively).

SW_{net} is both the strongest forcing ($+223.6 \text{ MJ}\cdot\text{m}^{-2}$) and also the most anomalous forcing compared to the ~~1980-present~~ 1980-present March average ($+72.0 \text{ MJ}\cdot\text{m}^{-2}$), followed by H_s ($+45.6 \text{ MJ}\cdot\text{m}^{-2}$, with a $+18.7$ anomaly) in the open site (Fig. 8a, Table 2). For the forest site, SW_{net} is also the strongest forcing ($+48.3 \text{ MJ}\cdot\text{m}^{-2}$) but not the most anomalous. The most anomalous forcing in the forest is LW_{net} , which is the second strongest forcing ($+27.8 \text{ MJ}\cdot\text{m}^{-2}$) with an anomaly of $+28.9 \text{ MJ}\cdot\text{m}^{-2}$ relative to historical conditions, indicating that ~~—~~ — for a typical year — LW_{net} is below-zero (and a source of snow cooling) rather than providing energy for heating snow under canopy. The lack of a ~~significant~~ significant LW_{net} anomaly in the open relative to the anomalous $LW \downarrow$ is in part explained by the compensating effects of outgoing $LW \uparrow$ — warmer snow radiates more energy ~~exponentially~~ exponentially (1).

The cumulative terms of panels Fig. 8a,d sum to the total Melt, with the small exception of the change in the initial snow cold-content (which is by definition zero when there is no snow) and is not depicted in the figure. The initial cold-content is small compared to the accumulated energy during the snow ablation period, and is on the order of $1 \text{ MJ}\cdot\text{m}^{-2}$ for the open site. The mass loss ~~assosiated~~ associated with Melt, combined with mass loss via sublimation, is equivalent to the total Δ SWE during the event. Only 3.1 cumulatively sublimated from the forest site compared with 0.4 mm from the open site ~~—~~ — which are less than 1% of the total ablation. In both sites this is less than half of the model estimated sublimation for typical March conditions (Table 2).

The above-canopy $SW \downarrow$ and $LW \downarrow$ (which are used as the top-boundary forcing for both open and canopy simulations) are also particularly anomalous during March 2026 (Fig. 8a). To understand these anomalies, we examine map-views and timeseries of the CERES EBAF CERES-EBAF product for the WUS and northern Sierra Nevada subset region (Fig. 9a-c). March of 2026 had the highest $SW \downarrow$ to the entire CERES record dating back to March of 2000 and is more than 1.5 standard deviations above average for the subset region. Nearly $40 \text{ W}\cdot\text{m}^{-2}$ of additional $SW \downarrow$ energy irradiated the land surface compared to average conditions. While less anomalous in terms of z-score, the $LW \downarrow$ anomaly is over $9 \text{ W}\cdot\text{m}^{-2}$ above average and the second-highest rate of monthly $LW \downarrow$ during the CERES EBAF CERES-EBAF record for this region.

The variations in the $SW \downarrow$ anomaly are almost entirely driven by variations in cloud radiative effect (which itself depends on cloud amount and optical properties). The same is not true for $LW \downarrow$, which also depends on other factors including water vapor and column T_{air} (Fig. 9c,d). RE_{cloud} values were the highest (i.e., the weakest rate of cooling) recorded by CERES EBAF CERES-EBAF, and correspondingly the RE_{cloud} LW effect was also the lowest on record, which together indicate that skies were anomalously free of optically thick clouds during this period. In other words, clouds typically provide about a $60 \text{ W}\cdot\text{m}^{-2}$ reduction in $SW \downarrow$ relative to clear skies in March, but only provided $20 \text{ W}\cdot\text{m}^{-2}$ during March. Correspondingly, no precipitation was observed during the month (Fig. 3c). With this in mind, it is even more remarkable that $LW \downarrow$ was above average for this region despite a clear-atmosphere, since clouds radiative as a nearly perfect gray-body at their base-temperature (Shupe & Intrieri, 2004).

In terms of energy, the anomalous increase $SW \downarrow$ due to the lack of optically thick clouds is equivalent to $93 \text{ MJ}\cdot\text{m}^{-2}$ integrated over the duration of snow cover in the open during March of 2026. Assuming 60% of that energy is reflected, this is equivalent to $37 \text{ MJ}\cdot\text{m}^{-2}$, or 110 mm of melt assuming that all of this energy is converted to melt (as opposed to sublimation or other terms of the energy balance). The monthly cadence of the CERES EBAF CERES-EBAF product does not allow us to discriminate if the peak of the anomaly corresponded with the early, cooler part of the month or the SEHW event later in the month.

4 Discussion

4.1 Sensible heat and sublimation do not scale linearly with T_{air}

The relatively small contributions of H_l and H_s despite anomalously high T_{air} can be explained by considering “offline” test cases assuming a fixed 0°C melting snow surface. Snow surface temperatures can never exceed this fundamental thermodynamic limit, leading to decoupling of the surface layer above the snow from the boundary layer air (Mott et al., 2013).

If a neutral condition is assumed without the effects of buoyant suppression, H_s over melting 0°C snow increases linearly as a function of T_{air} with a slope determined by w_{spd} (Fig. 10a). Accounting for buoyant suppression (using the same parameterization implemented in SUMMA; see TODO: supp) decreases H_s relative to the neutral assumption, and so much so that a moderate constant w_{spd} value of $5 \text{ m}\cdot\text{s}^{-1}$ yields yields H_s that actually decreases as a function of increasing T_{air} above approximately 4°C .

Fig. 10c depicts the T_{air} value that maximized H_s for a given w_{spd} . A w_{spd} of $2 \text{ m}\cdot\text{s}^{-1}$ achieves a maximum H_s when T_{air} is only a degree or two above freezing. A T_{air} of 25°C — slightly less than the daily maximum recorded at CSSL — requires a wind speed of over $10 \text{ m}\cdot\text{s}^{-1}$ to maximize H_s .

Evaporation and/or sublimation is likewise impacted by snow’s 0°C thermodynamic limit — the vapor pressure of pooling 0°C liquid water or ice on top of the

snowpack never exceeds 6.11 hPa to a reasonable approximation. Therefore, the vapor pressure of the air must be less than 6.11 hPa for sublimation or evaporation to occur. For a constant ~~RH~~-relative humidity (RH) of 20% (roughly the minimum observed during the March heatwave), H_l decreases rather than increases as a function of T_{air} over a 0°C snow regardless of w_{spd} (Fig. 10b,d). T_{air} greater than ~~24-25~~-24-25 $^\circ\text{C}$ at 20% RH adds energy and mass to the snow via condensation/deposition. On the other hand, sublimation is permitted across the full range of RH values as T_{air} approaches 0°C . In reality, though, RH is not independent of T_{air} , and generally decreases as T_{air} rises during the daytime due to water vapor conservation and dry-air entrainment from the upper levels of the boundary layer. RH values did reach these ~~low-levels~~-low levels for brief periods near noon, but were not sustained for long durations, explaining the relative lack of sublimation (less than ~~TODO: 1%~~ of the total snow mass loss during March). The effects of thermodynamic stability further reduce H_l as a function of T_{air} as well.

These arguments hold across elevation gradients with two additional considerations. All else being equal, H_s will ~~decreases~~-decrease with elevation as the density of dry air decreases (ρ in Supplemental Eq. 1). On the other hand, conditions that favor sublimation may ~~becomes~~-become more prevalent at higher elevations given the lapse rate of specific humidity and for snowpacks extending above the boundary layer.

The previous argument ~~assume that Monin-Obukov~~-assumes that Monin-Obukhov theory, which underpins turbulent exchange in nearly all land-models, is applicable. Mesoscale, terrain related flow features that generate turbulence from above the ground surface and provide additional mechanical turbulent mixing may invalidate the assumptions of this theory (Schwat et al., 2025). Likewise, the advection of sensible heat, such as from the forest edges and snow-free patches, may provide additional sensible/latent heat that is not accounted for (Harder et al., 2017). More work investigating turbulent exchange characteristics over snow during extreme heatwave conditions merits additional investigation.

4.2 Ablation and the relationship to overnight above freezing T_{air}

Whether the overnight T_{air} is above or below freezing is often considered an important factor explaining snow ablation, since refreezing overnight requires overcoming the energy required to re-break the ice-bonds the following day in order to generate liquid meltwater that may percolate out of the bottom of the snowpack. The snow does not instantly freeze when the two-meter T_{air} cools ~~to~~ 0°C , but rather depends on the energy balance of the snow which sheds heat via LW_{net} and H_s (if T_{air} is warmer than the snow layer).

We found that overnight T_{air} is correlated with the energy balance in many ways, but it is not alone a sufficient metric to inform rates of snow ablation or the snow energy balance writ-large (~~Fig. 11a,b~~)-during the March 2026 ablation period (~~Fig. 11a,b~~). Overnight refreezing is actually better correlated with overnight $LW \downarrow$ than T_{air} alone (R^2 of 0.13 vs. 0.41), even across the above/below freezing T_{air} threshold, and the same is true for overnight cold-content production (R^2 of 0.06 vs. 0.44). Overnight average T_{air} and $LW \downarrow$ similarly explain variance in next-day rates of melt. We caution that this analysis is not intended as a predictive model, keeping in mind these are all highly correlated variables. But this finding does highlight the importance of $LW \downarrow$ measurements and model products, since $LW \downarrow$ is better correlated with the overnight energy balance of the snow than T_{air} alone.

4.3 Limitations and caveats related to the influence of canopy

We only examined a limited case of vegetation canopy with a fixed LAI of 1.5, which corresponded well with CSSL observations (Fig. S4) and is representative of other forested regions of the Sierra Nevada (Van Gunst, 2012; Krogh et al., 2020). A full accounting

of the geographic pattern of vegetation/heatwave interactions likely ~~require~~requires considering additional processes such as canopy gap effects, shading from adjacent stands ~~and considering~~, and the full range of canopy types and parameters that are also aspect and slope dependent (Broxton et al., 2015).

Lundquist et al. (2013) presented a meta-analysis of snow climate studies and concluded that regions where ~~climatological~~climatological air temperatures exceed -1°C typically observe earlier melt out in forested areas, relative to open regions, and attribute the cause to the interactions related to the concepts expressed by our $\text{RF}_{\text{canopy}}$ analysis. They likewise conclude that forest snow is more sensitive to climate change in the Sierra Nevada than non-forested snow, given the temperature dependency of sub-canopy $LW \downarrow$ (Fig. 7e) but the lack of temperature dependency of $SW \downarrow$.

We showed that, while snow did ~~melt-out~~melt out earlier under canopy compared to the open, it did so at a lower rate, including during the period of the warmest SEHW conditions. The cause was the antecedent snow-interception rather than ~~and an~~an energy contribution, however, given the lower ablation rate. We have one important addition to Lundquist et al. (2013) concerning the climate sensitivity of snow under/out of vegetation ~~canopy~~canopy for this extreme event, T_{air} was anomalously high in addition to $SW \downarrow$ due to the lack of clouds (Fig. 9), on the order of $+40 \text{ W}\cdot\text{m}^{-2}$, meaning that shading effects by the canopy were also enhanced (Fig. 9). Future monthly timescale $SW \downarrow$ anomalies are ~~likely~~likely expected in the Sierra Nevada with warming. Some observational evidence is already pointing towards enhanced $SW \downarrow$ in ~~contentintal~~continental regions writ-large, including snow climates (Nyeki et al., 2019).

While we can conclude that forest provided resiliency to the SEHW event in this particular case (in terms of reducing the ~~the~~rate-of-ablation), more work is required to investigate the covariance of T_{air} and $SW \downarrow$ anomalies during SEHW events; warm but cloudy conditions may yield the opposite relationship. In addition, a key limitation of this work is the parameterization of the effects of leaf-litter on the α of the snow under canopy, which is not parameterized in SUMMA. More work is required to understand these interactions to establish optimal forest configurations for snow-resiliency against extreme heat conditions, as well as if the ~~climatological~~climatological relationships between mean T_{air} and forest snow processes (relative to open) hold for extreme events such as SEHW.

4.4 Future directions and relationship with SEHW events

The causes of extreme rates of snow ablation from SEHW in general, such as those reported in Rhoades et al. (2026) may occur for a variety of reasons that are not represented by the March 2026 event. In their work, a minimum of ~~3-days with above freezing~~3 days with above-freezing and $+6 \text{ K}$ ~~above-average above-average~~above-average T_{air} was used to classify SEHW events. We find that both the late February ROS event, and the late March dry and warm period, both qualify as a SEHW following the criteria presented in Rhoades et al. (2026), highlighting the variety of mechanisms that lead to anomalous snow ablation during periods of anomalous T_{air} . A complete accounting of the energy balance mechanisms during the wet-SEHW event in February is beyond the current scope, but ~~merit~~merits investigation.

We do find, however, that the timescales of anomalous ablation also mean that different mechanisms are important. For instance, it is well recognized that cold-content of the snow is vital for understanding event-scale ablation rates (Katz et al., 2023). While the February wet-SEHW and rain-on-snow event did deplete cold-content ~~TODO: cite supplement~~and ~~the~~reduce surface α , priming the snowpack for ablation, the initial cold content of the snow was not a large factor in explaining the anomalous monthly rate of ablation. For perspective, the initial cold-content was on the order of $0.8 \text{ MJ}\cdot\text{m}^{-2}$, whereas the cumulative energy directed towards melt during March was over $100 \text{ MJ}\cdot\text{m}^{-2}$, mean-

ing that the initial cold-content represented less than 1% of the energy directed towards ablation and is easily overcome. For some additional perspective, Jennings et al. (2018) also found that cold-content rarely exceeds $10 \text{ MJ}\cdot\text{m}^{-2}$ in cold, ~~continental~~ continental, alpine snowpacks. Both Jennings et al. (2018) and Seligman et al. (2014) also noted the timescale dependency of the relationship between cold-content and subsequent ablation.

But, the rates of overnight development of cold-content ~~does~~ do impact the daily rate ΔSWE , since the overnight cold-content production rate must be overcome the following day ~~prior~~ before melting can begin (Fig. 5f) and is as much as one-third of the following ~~days~~ day's rate of melt. The overnight ~~developement~~ development of cold-content is driven by LW_{net} and H_l in some limited cases, but is limited by the refreezing of melt-water overnight, which adds energy back to the snowpack (Fig. 5e).

Additional sensitivity tests that prescribed various snow initial conditions on 1 March, informed by historical snow states (Fig. S5), found that the accumulated amount of melt was largely insensitive to initial condition (or increased slightly for deeper snowpacks). However, deeper snowpacks were able to retain more liquid water, meaning the change in ΔSWE (which, again, is the totality of ice and retained liquid water in the snow) during the event was mitigated by as much as ~~TODO: 50 mm~~; in this way, higher antecedent SWE conditions are more resilient to snow ablation via their ability to retain more liquid water, not just initial cold-content of the snowpack.

This work suggests that analyzing the covariance between ~~heatwaves~~ heatwave $SW \downarrow$ and sustained $SW \downarrow$ anomalies (Fig. 9) ~~are important features~~ is an important feature to explore to understand SEHW impacts. Rough calculations suggest that as much as a third to a quarter of the anomalous ablation may be tied to the anomalous $SW \downarrow$ during the month of March ($40 \text{ W}\cdot\text{m}^{-2}$ higher than average). The relationship between anomalous T_{air} and $LW \downarrow$ is more complicated, however. It has previously been suggested that the relationship between $LW \downarrow$ and T_{air} may explain the efficacy of temperature index methods (Ohmura, 2001); this study does not necessarily support that conclusion, since $LW \downarrow$ anomalies are also associated with cloudy air-masses (Fig. 9). Nonetheless, examining the relationship between SEHW events and background precipitable water vapor merits investigation, given that it is a first order predictor of clear-sky $LW \downarrow$ (Brutsaert, 1975).

5 Conclusions

In this study, we used energy balance observations from CSSL, NRCS SNOTEL observations, snow energy balance modeling (SUMMA), atmospheric reanalyses (ERA5) and remote sensing products (~~CERES-EBAF~~ CERES-EBAF and SYN1Deg) to investigate interactions between anomalously warm T_{air} and snow ablation during the month of March 2026. The unprecedented rates of ablation were driven by several factors. The three-sigma March T_{air} anomaly also corresponded with the highest rate of $SW \downarrow$ observed during the CERES record (2000 to present) across the study region ($+20 \text{ W}\cdot\text{m}^{-2}$ above average for March) driven by a near-complete lack of mid-level ~~statiform~~ stratiform clouds. Combined with the complete absence of new snowfall, snow α declined rapidly.

The ~~higher-than-average~~ higher-than-average $SW \downarrow$ combined with ~~well-below-average~~ well-below-average snow albedo meant that net-shortwave rapidly ablated the snow in the ~~later~~ latter half of March. Vegetation emits longwave radiation at higher rates as the temperatures rose, but the added longwave to the sub-canopy snowpack was outweighed by shading effects, meaning that ~~both~~ snow ablates at a lower rate under canopy ~~relative~~ (XX mm/day) relative to the open site (XX mm/day). Overnight cold-content production influences variability in snow ablation. But, the extreme monthly rates of ablation are insensitive to reasonable variations in the initial thermodynamic state of the snowpack.

Sensible heating and sublimation were relatively minor factors, and the latter contributed less than 1% of the total snow ablation. Theoretical arguments show that, over 0°C snow, ~~that~~ sublimation is permitted only by an increasingly narrow range of atmospheric humidity values, which were not consistently maintained during the month of March. Likewise, atmospheric buoyancy effects mitigate sensible heating at high T_{air} and low winds. While most models make these assumptions, more work is needed to validate turbulent exchanges into snowpacks in complex terrain and in the vicinity of vegetation canopy where these assumptions may not hold. While snow-albedo parameterizations performed well against CSSL data, we propose that future observations should focus on albedo decay processes and grain-growth mechanisms during extreme heatwave events.

Heatwave events such as March 2026 are projected to increase with climate warming, necessitating efforts to better model and adapt water resource infrastructure and investigate ecosystem management for resiliency. Synoptic analyses of snow-eater heatwave events should focus on the co-variation of heatwave conditions with radiative anomalies and the dynamics that lead to the prolonged cloud-free conditions that contributed to the anomalous radiation conditions observed during March 2026.

Open Research Section

ERA5 atmospheric data and near-surface meteorological forcings for SUMMA were downloaded from the Copernicus data store ECMWF (2023). Geopotential height anomalies and standard deviations were precomputed and provided from the supplemental material of Lee et al. (2023). ~~SPIReS near-real-time~~ Snow Property Inversion from Remote Sensing (SPIReS) near-real-time data are publicly available from the National Snow and Ice Data ~~center~~ Center Snow Today page. The SUMMA model is publicly available on GitHub. The ~~VIIRS~~/Suomi-NPP VNP21A1D VIIRS Version 1 Land Surface Temperature product was processed and downloaded from Google Earth Engine **TODO: CITE GEE**.

Acknowledgments

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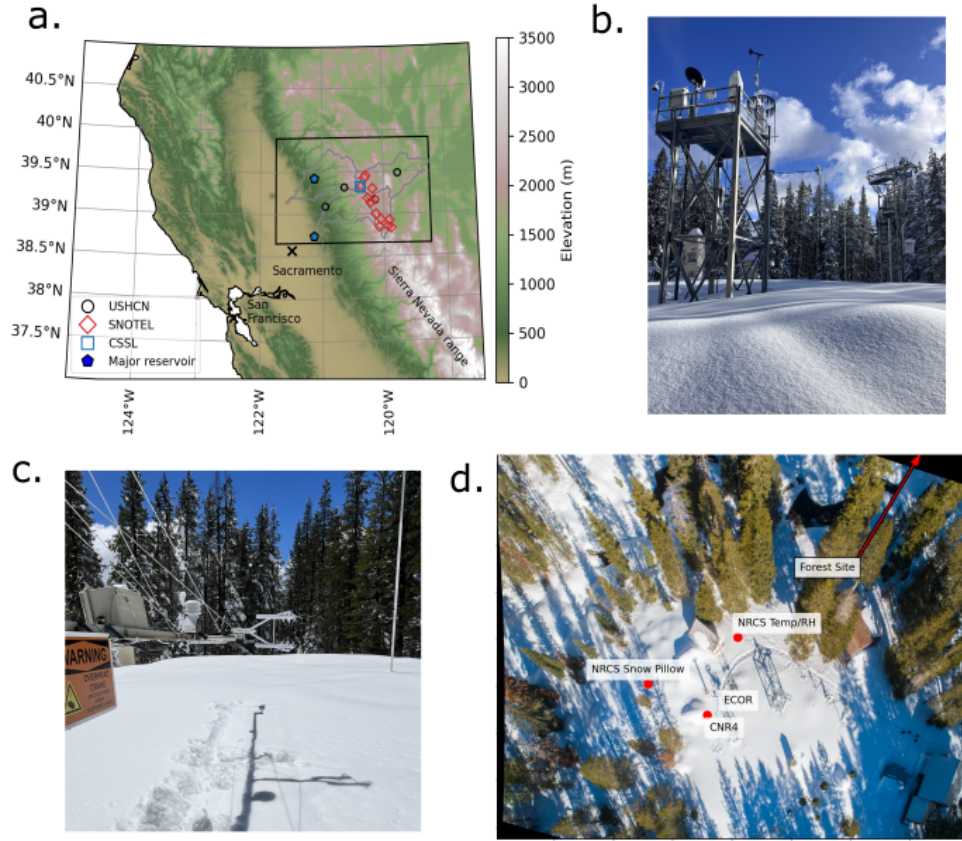


Figure 1. *TODO: change the font so it's the same as the other figures* (a) The northern California Sierra Nevada study region with terrain elevation. Outlines depict the Yuba, North Fork American, Lake Tahoe, and Truckee HUC8 watershed boundaries. Red open diamonds depict the location of NRCS SNOTEL sites, open black circles show the location of the four **USHCN** **USHCRN** stations in the region, and the open blue square depicts the CSSL energy balance site. (b) The CSSL meteorological tower in winter, (c) a close-up view of the Campbell CNR4 radiometer and IRGASON eddy covariance system, and (d) an aerial overview of the CSSL site. The location of the forest meteorological tower (not shown) is located just outside of the frame.

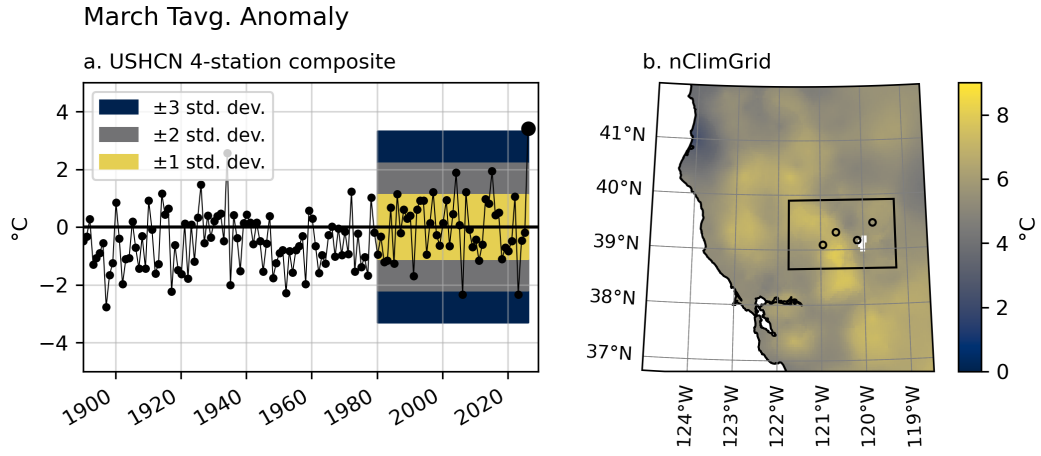


Figure 2. (a) Composite T_{air} anomalies from the four HCNv2 dataset stations in the greater Tahoe region study area for the month of March, from 1890 to 2026. Deviations from the 1980–2026 mean are depicted for one, two, and three standard deviation regions (shading). Anomalies are computed with respect to the 1980–2026 mean T_{air} . (b) The same, but for a map-view of the T_{air} anomaly of the gridded nClimGrid product.

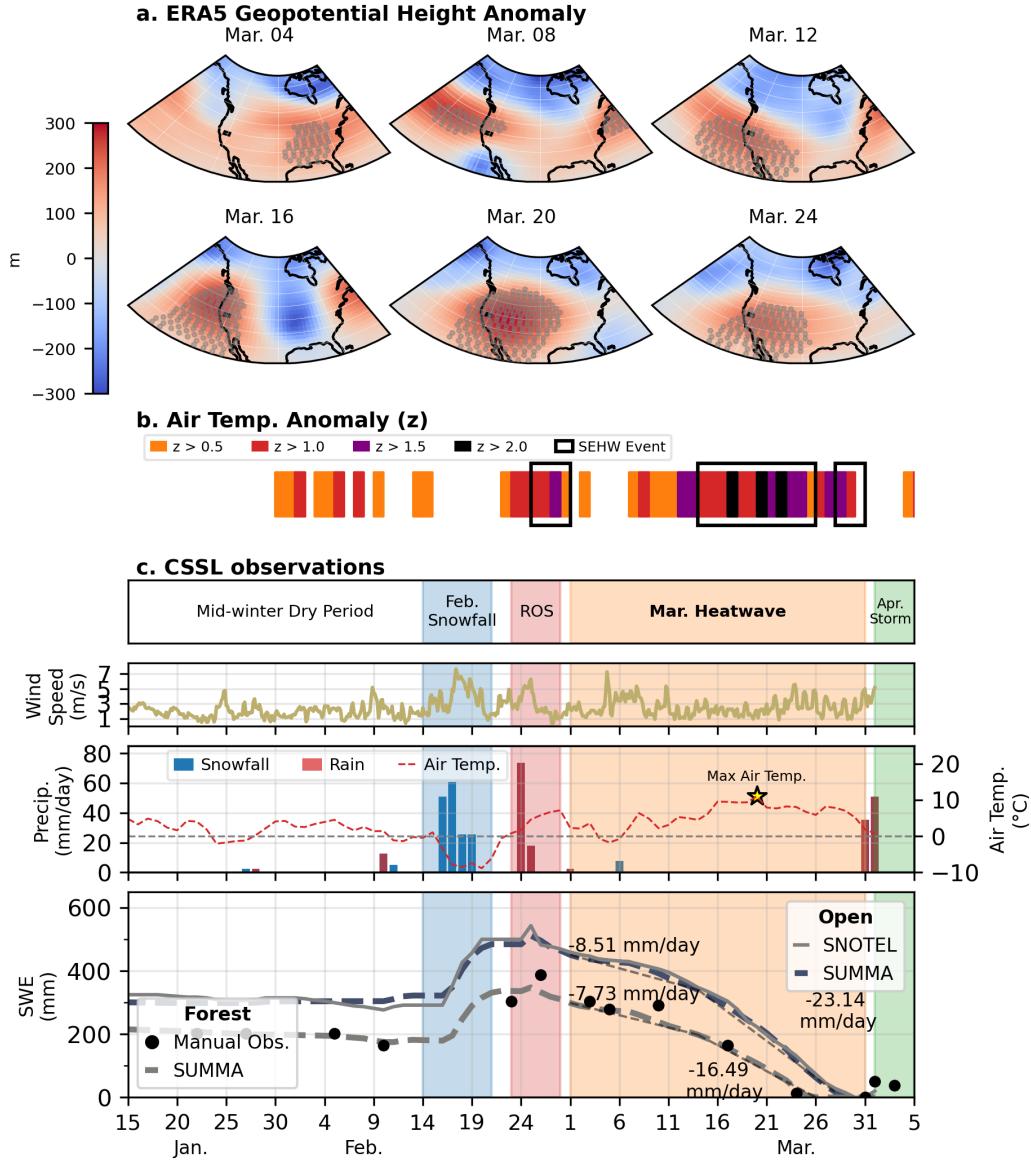


Figure 3. (a) The 500 ~~mb~~-hPa geopotential height anomaly during March 2026 heatwave from ERA5 across the WUS for six individual days. The study region is outlined using an overlain black rectangle. Gray stippling indicates regions with 500 ~~mb~~-hPa geopotential height anomalies greater than 2σ above the ~~TODO: 1981-2010~~ ~~TODO: 1981-2010~~ mean for the month of March. (b) depicts two meter T_{air} anomaly z-scores from the Tahoe City COOP station. Periods meeting the (Rhoades et al., 2026) SEHW criteria are outlined with a black box. (c) shows the time series of meteorological and snow conditions at CSSL, with different synoptic conditions highlighted by background shading (top-panel). Wind speed, precipitation with rain and snow differentiated, and two-meter T_{air} are depicted. The bottom panel depicts modeled (SUMMA) and observed SWE at the open and forest sites. Average rates of SWE loss are listed in the figure text with dashed lines superimposed on the SWE curves.

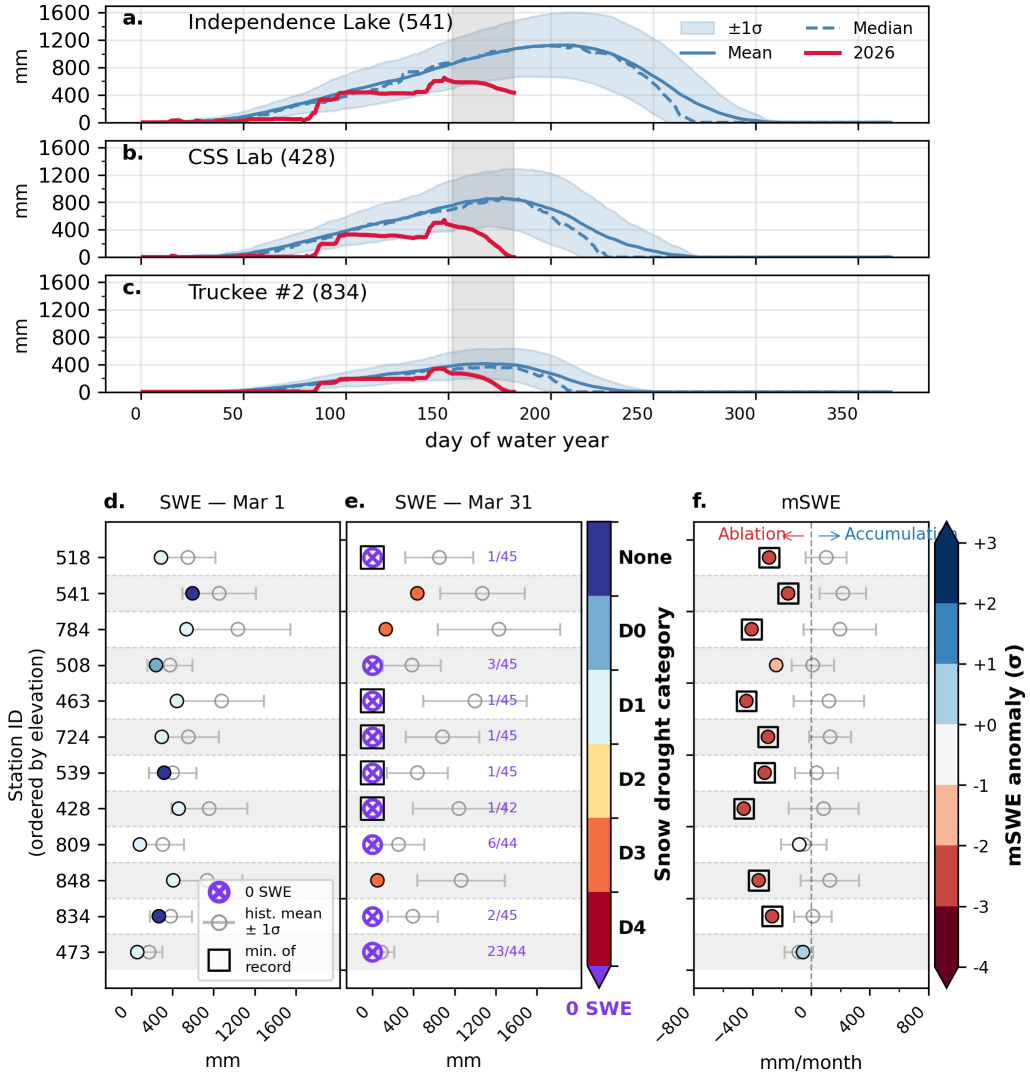


Figure 4. (a–c) A timeseries of SWE at three SNOTEL stations in the study region. The mean, median (dashed line), 1 standard-deviation (shaded region) and water year 2026 values are depicted. (d) SWE values on March 1st for the 12 SNOTEL sites in the study domain, labeled by their three-digit station code. (e) The same as (d), but for SWE on March 31. Color values correspond with the snow drought D-scale. A 0 mm of SWE value is depicted by a purple circle markersmarker. The purple text lists the number of 0 mm SWE occurrences observed on that date relative to the period of record for that site. (f) The SWE change between Mar 31 and Mar 1 (mSWE) compared with the historical record. Color values depict the deviation from the mean in units of standard deviation. Square markers bounding symbols indicates-indicate that the SWE or mSWE is the record minimum for the period of record.

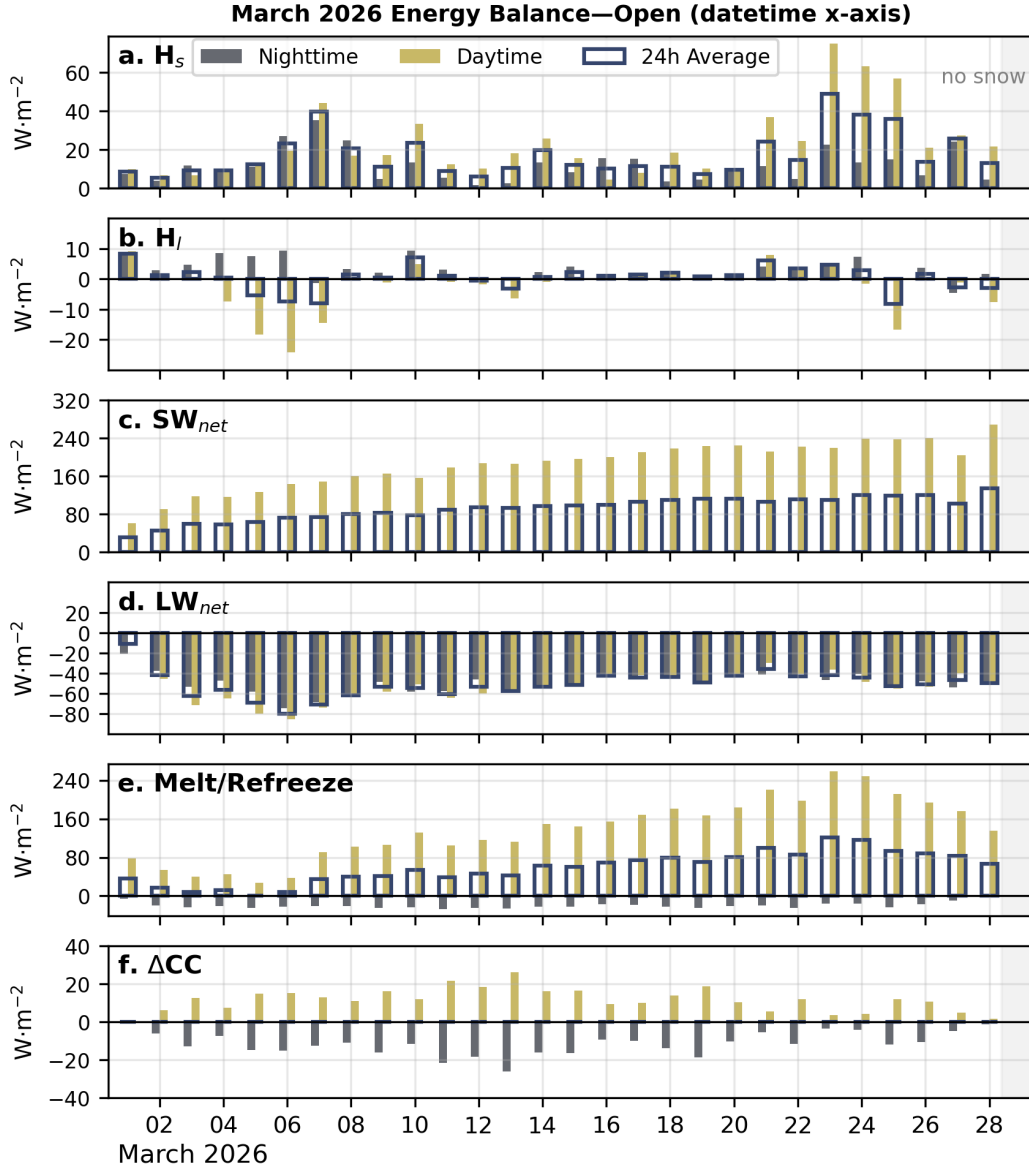


Figure 5. Daily average energy flux components including (a) H_s , (b) H_l , (c) SW_{net} , (d) LW_{net} , (e) snowpack Melt/Refreeze and (f) change in cold-content (ΔCC) during the March 2026 heatwave [at](#) the open site. Each day has three bars: the left-most yellow bar depicts the daytime flux, the right-most gray bar depicts the nighttime flux, and the outline spanning both bars depicts the 24-hour average flux. Please note that the y-scales are not equivalent across all of the figures.

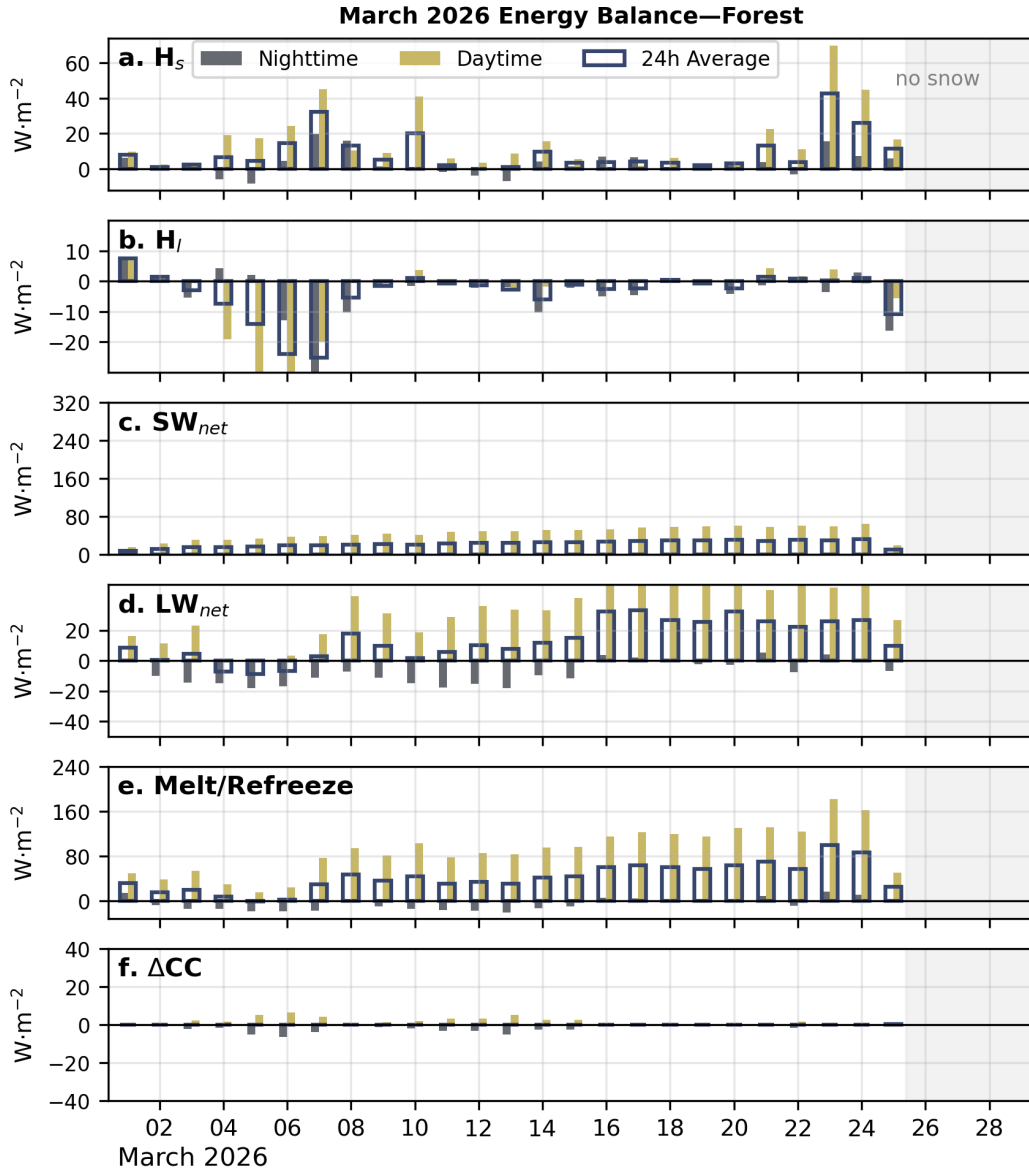


Figure 6. The same as Fig. 5 but for the forest site. Note that the y-axis limits and ticks are not the same as in (Fig. 5).

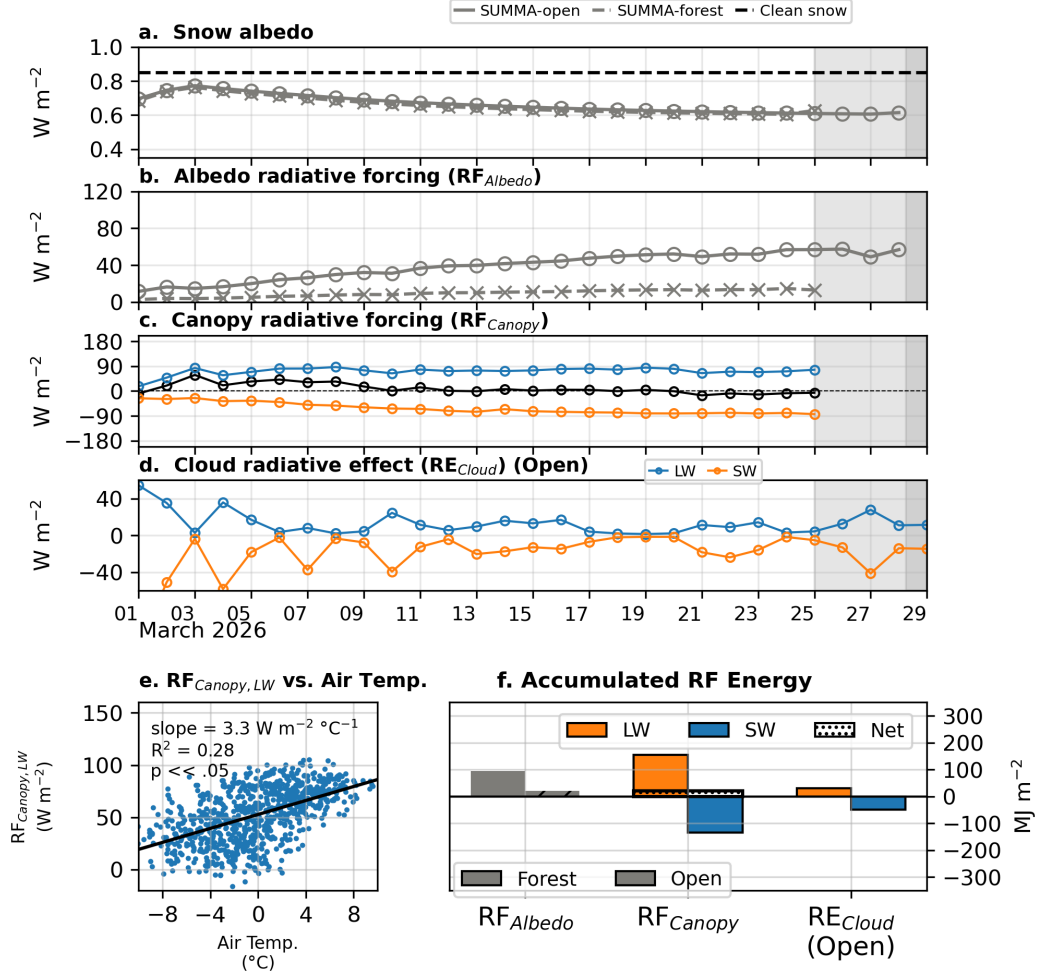


Figure 7. Timeseries of (a) α , (b) $\text{RF}_{\text{albedo}}$, (c) $\text{RF}_{\text{canopy}}$, (d) RE_{cloud} for March 2026. LW and SW components are depicted in (c) and (d). Net (LW + SW) components are also depicted in (c). (e) depicts the relationship between daily-average T_{air} and the LW component of $\text{RF}_{\text{canopy}}$ for all Marches in the SUMMA historical record (2000-2026). (f) depicts accumulated energy terms during the snow-covered duration of March.

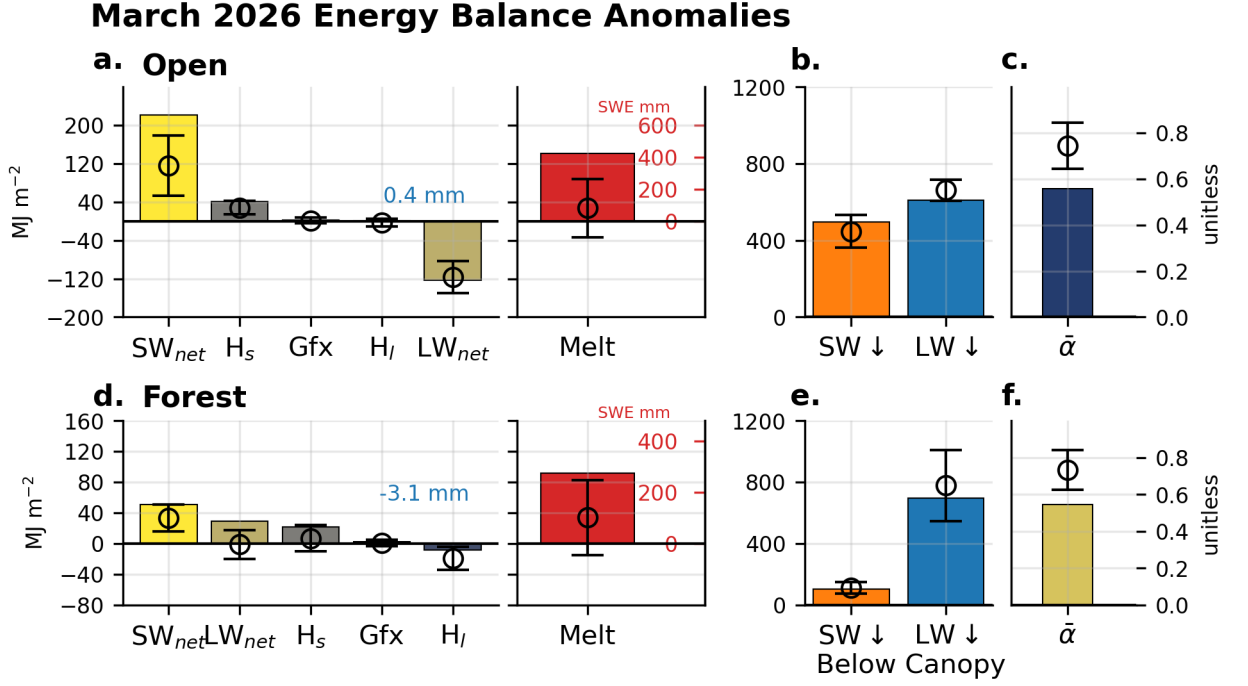


Figure 8. Accumulated snow energy fluxes (SW_{net} , LW_{net} , H_s , H_l , G), Melt, radiation forcings ($SW \downarrow$, $LW \downarrow$), and weighted snow $\bar{\alpha}$ for the open (a–c) and forest (d–f) sites for the duration of March, restricted to times when SWE is present. Open circles and whiskers show the averages and one-standard deviation from the March SUMMA historical period (2000–2026). Equivalent units of mm of melt are displayed in red-text on the right-hand side of the Melt panel. The under-canopy $SW \downarrow$ and $LW \downarrow$ are depicted in panel (e).

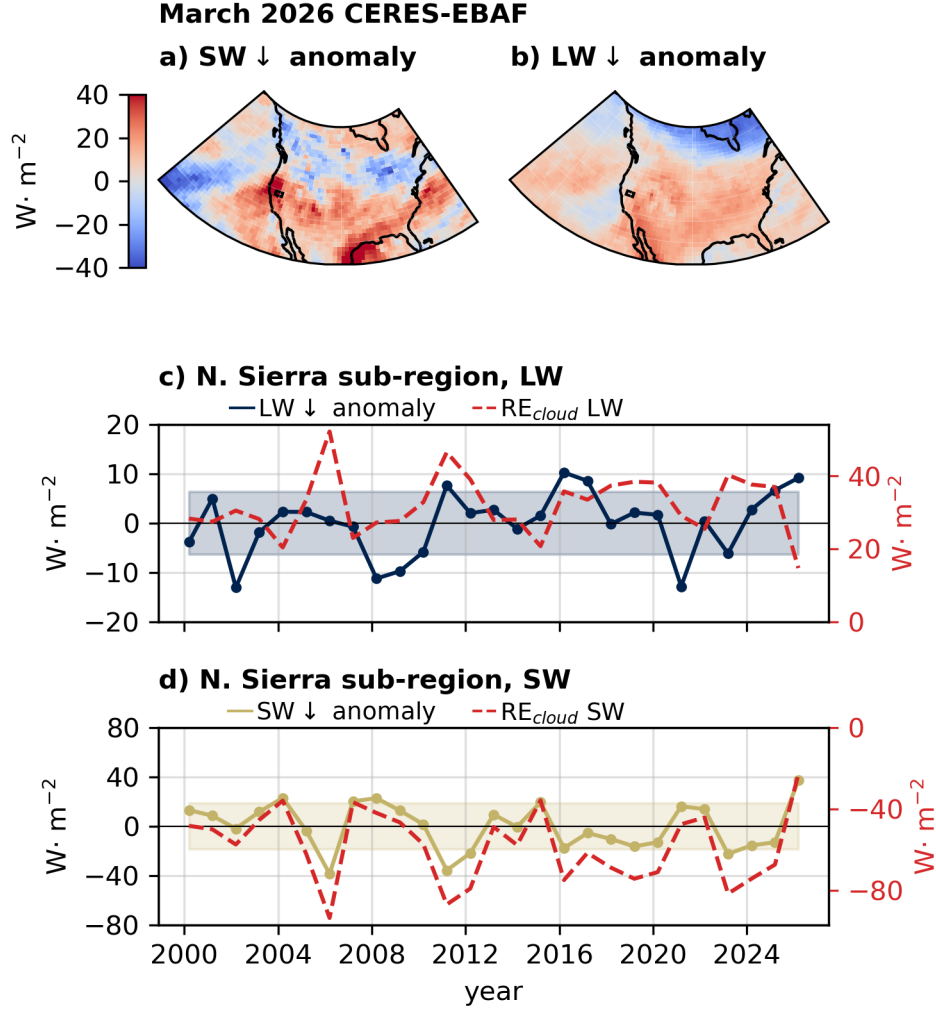


Figure 9. March $SW \downarrow$ (a) and $LW \downarrow$ (b) anomalies (relative to the March 2000 to present record) from the ~~CERES-EBAF~~ CERES-EBAF 1x1 degree resolution product. (c) and (d) ~~depicts~~ depict the timeseries of anomalies averaged over the Sierra Nevada study region. RE_{cloud} amount is depicted with a twin y-axis, with units displayed on the ~~righthand~~ right-hand side in red text. The shaded region in each plot depicts one standard deviation of the anomaly.

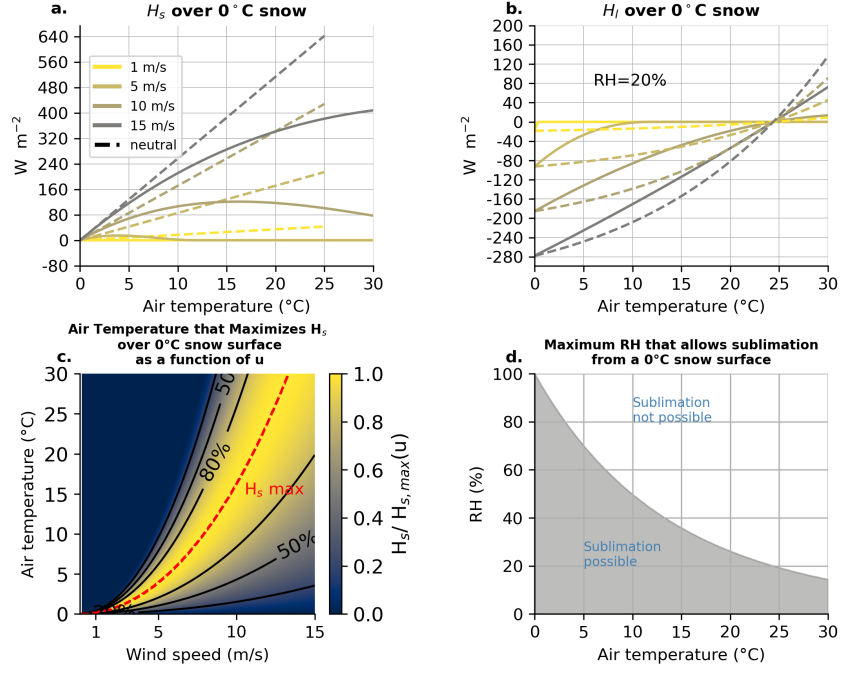


Figure 10. (a) Idealized sensible heat fluxes (H_s) into a melting snowpack as a function of T_{air} for four wind speeds (1, 5, 10, and 15 $m\ s^{-1}$) at a 12 m height for open-conditions and a snow surface roughness of $z_0=0.002\ m$. Dashed lines show neutral-stability bulk transfer; solid lines show the same formulation with a stability correction applied as implemented in SUMMA. (b) the same as (a), but for H_l with a fixed RH of 20%. (c) The T_{air} that yields the maximum H_s into a 0 °C snowpack after correcting for stability effects (dashed red line) for a given wind speed (u). Contours depict the fraction of the maximum H_s for that wind speed value. (d) The thermodynamic limitation of sublimation as a function of T_{air} , depicting the maximum RH that permits sublimation from a 0 °C snowpack.

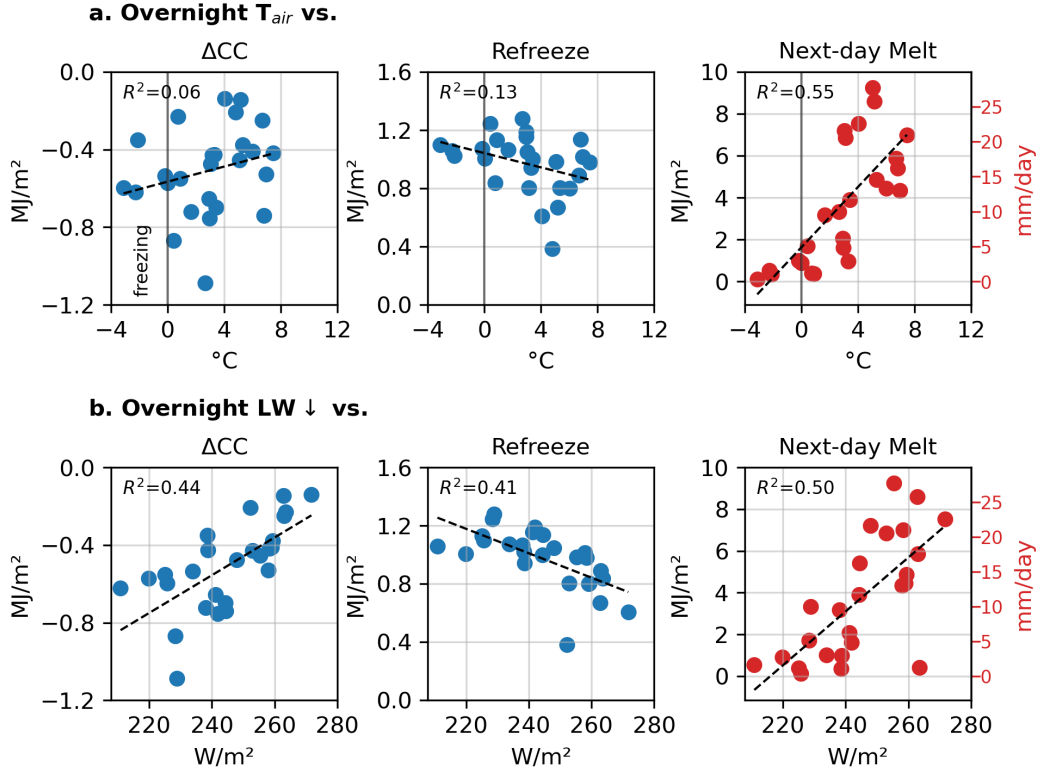


Figure 11. The predictive power of overnight average T_{air} compared against overnight average LW_{net} for ΔCC , refreezing, and total melt during the next-day. R^2 values and linear-regressions (dashed line) are depicted.

Table 2. Data displayed in Figure 8 presented as a table. Equivalent units of mm of SWE for H_l and Melt are presented in parentheses. The March 2026 change in SWE δ SWE and historical mean and standard deviation are also presented.

		SW_{net}	LW_{net}	H_s	H_l	G	Melt	Δ SWE
Forest	WY2026	50.70	28.80	21.20	-8.80 (-3.1)	2.20	91.50 (274.0)	-306.20 —
	Mean	33.00	-1.30	6.90	-19.50 (-6.9)	0.70	33.70 (100.9)	140.10 —
	Std	17.90	19.20	17.50	15.50 (5.5)	4.20	49.60 (148.5)	424.30 —
Open	WY2026	221.20	-123.20	41.40	1.10 (0.4)	2.80	140.80 (421.6)	-461.10 —
	Mean	116.10	-116.90	27.90 —	-2.90 (-1.0)	1.50	27.40 (82.0)	185.30 —
	Std	64.40	34.20	14.30	8.50 (3.0)	6.50	61.90 (185.3)	469.00 —

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